Product Requirements Document (PRD)

Overview

MediFlow AI is a working prototype of a clinical operations copilot for a fictional outpatient clinic in Ghana (MediFlow Clinic). It runs entirely on synthetic data and shows, end to end, how LLMs, RAG, a predictive model and an AI agent work together inside one product, with a human approving every step that matters.

Product goal

Show a hiring team, in one 3-minute demo, that the builder can design and ship production-shaped AI features for healthcare: structured LLM output, grounded RAG with citations, an explained ML model, a tool-using agent with approval gates, and measured results.

The demo story: one patient, Ama, goes from consultation to signed note, guideline check, care plan, follow-up booked, plain-English instructions sent, reminder sent, and NHIS claim checked and paid.

Users

Role	What they do in MediFlow	Main screens
Doctor	Generates and approves clinical notes; checks prescriptions against guidelines	Consultations, Note review
Nurse	Approves care plans and after-visit messages; answers clinical questions from patients	Care plans, Outbox, Nurse inbox, Alerts
Front desk	Watches upcoming appointments and risk; marks attendance	Schedule
Billing officer	Reviews, fixes and submits NHIS claims	Claims queue, Claim detail
Patient (simulated)	Receives reminders and instructions; replies	Patient chat
Admin (the builder, in demos)	Ingests documents, runs jobs, resets demo data, views results	Admin, Results dashboard

Features

ID	Feature	Description	Expected behaviour
F1	Scribe (Module 1)	Turns a consultation transcript into a structured clinical note	Masked text goes to the LLM; a 13-field note comes back; the doctor edits and approves
F2	Guideline check (Module 2)	Compares the prescription with clinic guidelines	Each drug marked matches / differs, with citations, before approval
F3	Ask the guidelines (Module 2)	Chat over 3 clinic documents	Cited answers; "not covered" when the documents don't say
F4	No-show risk (Module 3)	Scores each appointment's risk of being missed	High / Medium / Low band with a one-sentence reason

ID	Feature	Description	Expected behaviour
F5	Patient messenger (Module 4)	Reminders, after-visit instructions, replies to patients	Templates for reminders; nurse-approved instructions; clinical questions go to a nurse; emergencies caught first
F6	Care coordinator (Module 5)	Agent that builds each patient's care plan	Calls modules as tools; proposes actions; nothing runs until a nurse approves
F7	Claims checker (Module 6)	Builds and checks NHIS claims	Rule and LLM flags, red / amber / green; officer submits to simulated NHIS
F8	Results dashboard	One page of evaluation results across modules	KPIs from every eval script, latest run, with model and prompt versions
F9	Demo controls	Make the demo repeatable	Reset data, "run daily job for date", role switcher

Out of scope for this version

- Real patient data, real NHIS submission, and real WhatsApp or SMS (the chat is simulated).
- Audio transcription (transcripts are the input).
- Languages other than English.
- Readmission prediction and claim-rejection ML.
- Real authentication (a role switcher stands in for login).

Success metrics

Area	Metric	Target
Scribe	Diagnosis and code match the answer key	⩾ 8 of 10 transcripts
RAG	Retrieval hit rate / refusal accuracy	⩾ 90% / 5 of 5
Risk model	AUC; beats logistic regression baseline	⩾ 0.75; yes
Messenger	Emergency catch rate; clinical questions routed to a nurse	100%; 100%

Area	Metric	Target
Coordinator	Actions executed without approval	0
Claims	Rejection rate before → after	~30% → ⩽ 8%
Demo	Ama's full journey runs without manual fixes	Under 3 minutes